Chain Rule

Given a function $f : \mathbb{R}^m \to \mathbb{R}^n$

At some point $x_0 \in \mathbb{R}^m$, approximate f by a linear map $Df(x_0) \in \mathcal{L}(\mathbb{R}^m \to \mathbb{R}^n)$ or $\mathbb{R}^{n\times m}$

$$\underbrace{f(x)}_{\mathbb{R}^n} = \underbrace{f(x_0)}_{\mathbb{R}^n} + \underbrace{Df(x_0)}_{\mathbb{R}^{n\times m}}\underbrace{(x-x_0)}_{\mathbb{R}^m} + \underbrace{o(x-x_0)}_{\mathbb{R}^n}$$

this term is strictly smaller than $(x-x_0)$ when $(x-x_0) \to 0$

$$\lim_{x_1 \to x} \frac{\|o(x_1-x_0)\|}{\|x_1-x_0\|} = 0$$

Total Derivative

$$D : \overbrace{(\mathbb{R}^m \to \mathbb{R}^n) \to \mathbb{R}^m \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)}^{\text{function currying notation}}$$

- $A \to B$: set of all functions $f : A \to B$
- $A \to B \to C$: equivalent to $A \to (B \to C)$ or $A \times B \to C$

loosely speaking,

D is an operator acting on the set of all functions from $\mathbb{R}^m$ to $\mathbb{R}^n$ producing a linear map on every points in $\mathbb{R}^n$.

$\to$ not strict, not all functions are differtiable
not all differtiable functions are differentiable everywhere

Chain Rule

$f : \mathbb{R}^m \to \mathbb{R}^k$, $g : \mathbb{R}^n \to \mathbb{R}^m$, $f \circ g : \mathbb{R}^n \to \mathbb{R}^k$

At $x_0 \in \mathbb{R}^n$, given $Dg(x_0)$ and $Df(g(x_0))$
find $D(f\circ g)(x_0)$

we have

$$(f\circ g)(x) = (f\circ g)(x_0) + D(f\circ g)(x_0)(x-x_0) + o(x-x_0)$$

or

$$(f\circ g)(x) = f(g(x))$$

Let $y = g(x)$, $y_0 = g(x_0)$

$$(f\circ g)(x) = f(y)$$
$$= f(y_0) + Df(y_0)(y-y_0) + o(y-y_0)$$

furthermore

$$y - y_0 = g(x) - g(x_0)$$
$$= Dg(x_0)(x-x_0) + o(x-x_0)$$

Hence,

$$(f\circ g)(x) = f(y_0) + \underbrace{Df(y_0)}_{\in \mathcal{L}(\mathbb{R}^m \to \mathbb{R}^k)}\Big[\underbrace{\overbrace{Dg(x_0)}^{\mathcal{L}(\mathbb{R}^n \to \mathbb{R}^m)}\overbrace{(x-x_0)}^{\mathbb{R}^n} + o(x-x_0)}_{\mathbb{R}^m}\Big] + o(y-y_0)$$

$$= (f\circ g)(x_0) + \underbrace{Df(g(x_0))}_{\mathcal{L}(\mathbb{R}^m \to \mathbb{R}^k)}\underbrace{Dg(x_0)}_{\mathcal{L}(\mathbb{R}^n \to \mathbb{R}^m)}\underbrace{(x-x_0)}_{\mathbb{R}^n} + \overbrace{\Big(\underbrace{Df(g(x_0))}_{\mathcal{L}(\mathbb{R}^m \to \mathbb{R}^k)}\underbrace{(o(x-x_0))}_{\mathbb{R}^m} + \underbrace{o(y-y_0)}_{\mathbb{R}^k}\Big)}^{o(x-x_0)}$$

$\Big\}$ $Df(g(x_0))$ is linear

by assuming $\|Df(g(x_0))\| < \infty$ and $\|Dg(x_0)\| < \infty$

- $$\lim_{x_1 \to x} \frac{\|o(g(x_1)-g(x_0))\|}{\|x_1-x_0\|} = \lim \frac{\|o(g(x_1)-g(x_0))\|}{\|g(x_1)-g(x_0)\|} \cdot \underbrace{\frac{\|g(x_1)-g(x_0)\|}{\|x_1-x_0\|}}$$

$$\frac{\|Dg(x_0)(x_1-x_0) + o(x_1-x_0)\|}{\|x_1-x_0\|} \leq \|Dg(x_0)\| \cdot \|x_1-x_0\| + \|o(x_1-x_0)\|$$

$$\leq \|D(g(x_0))\| + \frac{\|o(x_1-x_0)\|}{\|x_1-x_0\|}$$

$$\leq \frac{\|o(g(x_1)-g(x_0))\|}{\|g(x_1)-g(x_0)\|}\left(\|D(y(x_0))\| + \frac{\|o(x_1-x_0)\|}{\|x_1-x_0\|}\right)$$

- $$\lim_{x_1 \to x} \frac{\|Df(g(x_0))(o(x_1-x_0))\|}{\|x_1-x_0\|}$$

$$\leq \|Df(g(x_0))\| \, \|o(x_1-x_0)\|$$

$$\|Df(g(x_0))\| \frac{\|o(x_1-x_0)\|}{\|x_1-x_0\|}$$

Chain Rule

$$D(f\circ g)(x) = Df(g(x))\, Dg(x)$$